Synergistic Integration of Scilab-Xcos Dynamic Modeling with SurfactantKit Molecular Thermodynamics: A Roadmap for Novel Discoveries in Interfacial Science

Short Title: Scilab-Xcos and SurfactantKit Research Synergy

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Abstract

A central challenge in contemporary colloid and interface science is the profound computational divide between molecular-scale equilibrium thermodynamics and continuum-scale process dynamics. While existing computational toolkits excel at calculating equilibrium properties such as critical micelle concentration, packing parameters, and static adsorption isotherms, they fail to capture the transient, non-equilibrium phenomena that govern real-world applications including microfluidic emulsification, dynamic foam drainage, enhanced oil recovery, and pulmonary surfactant mechanics. In this work, we propose a synergistic integration between SurfactantKit, an open-source Python library for molecular and thermodynamic interfacial properties, and Scilab-Xcos, a high-performance open-source numerical environment and dynamic block-diagram simulator. By coupling SurfactantKit's microscopic molecular descriptors and adsorption equations of state with the stiff differential-algebraic equation solvers of Scilab-Xcos, researchers can simulate dynamic Marangoni stresses, micellar relaxation kinetics, and transient interfacial dilation across millisecond-to-minute timescales. We present four specific, high-impact research proposals designed for publication in premier peer-reviewed journals (such as Journal of Colloid and Interface Science, Langmuir, Chemical Engineering Science, and Computers & Chemical Engineering): (1) nonlinear large-amplitude oscillatory dilatational rheology and harmonic distortion; (2) dynamic Marangoni tip-streaming and droplet breakup in microfluidic junctions; (3) acoustic and mechanical resonance in thin liquid film foam drainage; and (4) an open-source, reproducible

benchmark for dynamic colloid engineering. Furthermore, an executable mathematical formulation and Scilab prototype are provided, establishing a clear operational roadmap for generating novel, publishable scientific breakthroughs without reliance on proprietary simulation suites.

Keywords: SurfactantKit, Scilab, Xcos, dynamic surface tension, interfacial rheology, Marangoni stress, microfluidics, foam drainage.

1. Introduction

Surfactant formulations represent the cornerstone of modern chemical manufacturing, personal care formulations, pharmaceutical drug delivery systems, enhanced oil recovery (EOR), and mineral flotation [1,2]. Over several decades, academic and industrial research has relied predominantly on equilibrium characterization techniques, such as equilibrium surface tension isotherms, critical micelle concentration (CMC) measurements, and static contact angle measurements [3-5]. In parallel, computational colloid chemistry has bifurcated into two disparate extremes: atomistic molecular dynamics (MD) simulations on the one hand, and classical computational fluid dynamics (CFD) on the other [6,7].

Atomistic and coarse-grained molecular dynamics simulations provide invaluable microscopic insights into micellar aggregation numbers, headgroup hydration shells, and bilayer self-assembly [8,9]. However, MD is fundamentally constrained by spatial and temporal boundaries, rarely capturing phenomena beyond 10-100 nanoseconds and tens of nanometers, which renders it computationally intractable for macroscopic transport processes. Conversely, conventional CFD and process engineering simulators treat interfacial tension as an instantaneous, scalar constant or rely on crude empirical correlations that ignore molecular relaxation phenomena [10,11]. Real-world interfacial operations—such as droplet breakup in high-shear microfluidic mixers, rapid bubble expansion during lung

respiration, and transient foam film rupture—occur in non-equilibrium regimes spanning milliseconds to tens of seconds [12-14].

To bridge this mesoscale continuum divide, an open-source dynamic co-simulation architecture is required. Recently, the open-source package SurfactantKit has emerged to compute molecular descriptors, critical packing parameters, mixed-micelle non-ideality, and Ward-Tordai adsorption kinetics directly from chemical structure [15]. Simultaneously, Scilab and its dynamic block-diagram environment Xcos provide industrial-grade, open-source differential-algebraic equation (DAE) solvers, event-driven hybrid simulation engines, and Modelica-based physical modeling capabilities [16]. In this paper, we systematically evaluate whether coupling Scilab-Xcos with SurfactantKit can generate genuinely novel scientific findings capable of supporting premium Q1 publications, outline four high-impact investigation proposals, and provide a concrete implementation blueprint.

2. Computational Architecture and Bridging Framework

The proposed computational synergy couples molecular-level thermodynamics with dynamic continuum state-space integration. Figure 1 illustrates the bidirectional communication between the microscopic property engine (SurfactantKit) and the macroscopic dynamic simulator (Scilab-Xcos).

2.1. Thermodynamic Parameter Engine: SurfactantKit

SurfactantKit is structured as a modular Python framework designed for theoretical and applied surfactant science. Its core modules evaluate molecular structure from simplified molecular-input line-entry system (SMILES) strings, determining tail volume (v), effective extended chain length (lc), and polar headgroup area (a0) to predict the critical packing parameter (CPP = v / (a0 * lc)) [17]. Furthermore, the library incorporates established thermodynamic formulations, including the Langmuir-Szyszkowski, Frumkin, and Temkin adsorption isotherms, Rubingh-Rosen regular solution theory for

multicomponent mixed micelles, hydrophilic-lipophilic difference (HLD-NAC) phase behavior coordinates, and the Ward-Tordai diffusion-controlled dynamic adsorption integral [18-20].

2.2. Dynamic State-Space Integrator: Scilab and Xcos

Scilab provides a matrix-based computational environment equipped with the SUNDIALS solver suite, specifically CVode for stiff ordinary differential equations (ODEs) and IDA for differential-algebraic equations (DAEs) [21]. Xcos extends Scilab with a graphical, multi-domain dynamic modeling canvas analogous to Simulink. Within Xcos, physical processes are represented as interconnected blocks that support continuous integration, discrete-time sampling, zero-crossing event detection, and Modelica physical components [22]. This makes Xcos exceptionally well-suited for solving boundary-layer mass transfer equations, transient interfacial area deformations, dynamic Marangoni stress balances, and thin-film drainage hydrodynamics.

2.3. Bidirectional Coupling Protocols and Data Flow

To achieve seamless co-simulation between Python-based SurfactantKit and Scilab-Xcos, three integration architectures are established: (1) an in-process direct C-API or Python gateway, where Scilab's built-in Python integration executes SurfactantKit routines directly during runtime; (2) a high-speed inter-process communication (IPC) protocol using shared memory or local loopback sockets exchanging JSON-serialized state vectors; and (3) a Functional Mock-up Interface (FMI) standard where SurfactantKit models are compiled into Functional Mock-up Units (FMUs) ingested by Xcos [23]. Table 1 contrasts this open-source architecture with conventional proprietary alternatives.

Table 1. Comparative architectural analysis of computational modeling frameworks for dynamic surfactant systems.

Platform Architecture	**Thermodynamic Rigor**	**Dynamic Solvers**	**Licensing & Cost**	**Reproducibility & Openness**
MATLAB / Simulink + In-house Code	Proprietary scripts	ODE15s / Stiff DAE	Expensive commercial license	Restricted access
Aspen Custom Modeler / COMSOL	Built-in EOS / CFD	Finite element / DAE	Heavy enterprise tier	Proprietary binary

Pure Python (Scipy / Cantera)	High (SurfactantKit)	Basic SciPy ODE / IVP	Free & Open Source	High reproducibility
Scilab-Xcos + SurfactantKit	High (SurfactantKit)	SUNDIALS CVode / IDA / Xcos	Free & Open Source	Complete community access

3. Four Breakthrough Research Proposals for Premium Publications

By merging SurfactantKit's molecular thermodynamics with Xcos's continuum dynamic solvers, researchers can tackle complex interfacial problems that currently remain unsolved due to software bifurcation. Below, four concrete research directions are detailed.

3.1. Nonlinear Large-Amplitude Oscillatory Dilatational (LAOD) Rheology and Harmonic Marangoni Stress

In interfacial dilatational rheometry, sinusoidal area oscillations are imposed on a pendant drop or bubble (A(t) = A0 * [1 + epsilon * sin(omega * t)]). Current literature predominantly evaluates the dynamic interfacial tension through linear viscoelastic theory, as formulated by Lucassen and van den Tempel (LVDT), which is strictly valid only in the limit of infinitesimal strain amplitudes (epsilon < 0.05) [24,25]. However, industrial atomization, high-speed coating, and physiological alveolar expansion involve large cyclic deformations (epsilon = 0.20-0.50) where linear assumptions collapse.

Under large-amplitude oscillations, nonlinear surface coverage, cohesive Frumkin intermolecular lateral interactions (parameter beta), and finite micellar breakdown kinetics (Aniansson-Wall relaxation times tau1 and tau2) generate asymmetric Marangoni stress profiles [26,27]. By coupling SurfactantKit's non-ideal equation of state with Xcos's nonlinear DAE solver, one can model the exact boundary-layer diffusion flux coupled with micellar source/sink terms (dM/dt = -(M - Meq) / tau2). The resulting dynamic surface tension response exhibits higher-order Fourier harmonics (2*omega, 3*omega). Discovering how specific surfactant molecular structures (e.g., gemini surfactants vs. single-chain ethoxylates) generate distinct harmonic fingerprints provides a novel scientific breakthrough directly suitable for Journal of Colloid and Interface Science (JCIS) or Langmuir.

3.2. Dynamic Marangoni Tip-Streaming and Droplet Breakup in Microfluidic Cross-Flow Junctions

Microfluidic droplet generation in flow-focusing geometries operates at millisecond capillary timescales. When an aqueous surfactant solution is sheared by an immiscible continuous oil phase, fresh interface is rapidly generated at the droplet tip (dA/dt >> 0) [28,29]. Because surfactant diffusion and adsorption require finite time, dynamic surface tension (gamma(t)) at the protruding neck is substantially higher than the equilibrium surface tension (gamma_eq). This gradient creates intense Marangoni stresses (tau_M = grad(gamma)) that act opposite to the fluid drag, either delaying pinch-off or triggering tip-streaming of sub-micron satellite droplets [30].

Using Xcos, researchers can construct a lumped-parameter hydrodynamic network simulating the coupled continuous-phase pressure drop, neck contraction velocity, and time-dependent capillary number (Ca(t) = mu * v / gamma(t)), with gamma(t) computed in real-time by SurfactantKit's numerical Ward-Tordai kernel. This enables the discovery of a universal regime map predicting the transition from stable dripping to irregular jetting as a function of the surfactant's characteristic adsorption Damkohler number (Da = t_hydro / t_ads). This investigation directly addresses the core scope of Chemical Engineering Science or Soft Matter.

3.3. Acoustic and Mechanical Resonance in Dynamic Foam Lamella Drainage and Rupture

A critical problem in foam science is understanding why specific mixed surfactant formulations (e.g., anionic/zwitterionic or catanionic pairs) resist mechanical vibration and acoustic noise during industrial transport or firefighting operations [31,32]. Classical foam drainage models rely on steady-state Plateau border drainage equations, neglecting the oscillatory perturbation of the liquid film thickness [33].

SurfactantKit calculates the disjoining pressure isotherm Pi(h), incorporating electrostatic double-layer repulsion via the nonlinear Poisson-Boltzmann equation, van der Waals attraction, and short-range steric hydration, alongside the Gibbs surface elasticity (EG = -d(gamma) / d(ln(Gamma)))

[34]. In Xcos, the thin-film Reynolds drainage equation (dh/dt) is integrated under external sinusoidal pressure fluctuations. Coupling the dynamic surface tension recovery to film drainage reveals subharmonic resonance modes: at critical vibrational frequencies, localized Marangoni backflow fails to heal the thinned region, triggering catastrophic film rupture at a critical thickness (hcrit). Publishing these frequency-dependent foam rupture thresholds offers a high-impact contribution for Colloids and Surfaces A: Physicochemical and Engineering Aspects or JCIS.

3.4. An Open-Source Framework for Multiscale Colloid Process Simulation

Beyond individual mechanistic discoveries, the literature urgently requires a benchmarked, open-source software methodology capable of replacing closed commercial simulation suites [35]. By formally publishing the Scilab-Xcos and SurfactantKit co-simulation architecture, validating its numerical accuracy against standard analytical benchmarks (such as Ward-Tordai asymptotic limits) and experimental tensiometry datasets, this contribution represents a premier software and modeling paper targeted at Computers & Chemical Engineering or Journal of Chemical Information and Modeling.

Table 2. Strategic research proposal roadmap, journal publication targets, novelty metrics, and validation data sources.

Project Focus	**Primary Target Journal**	**Target Impact Factor**	**Core Scientific Novelty**	**Primary Validation Source**
Nonlinear LAOD Interfacial Rheology	Journal of Colloid and Interface Science	9.9	Higher harmonic generation & micelle relaxation filtering	Pulsating drop tensiometry (PAT-1M)
Dynamic Microfluidic Droplet Breakup	Chemical Engineering Science	4.7	Dynamic Ca(t) mapping & Marangoni neck stabilization	High-speed microfluidic video datasets
Acoustic Resonance in Foam Drainage	Colloids and Surfaces A	5.2	Subharmonic film rupture bifurcation & vibration resilience	Foam scan & thin-film balance data
Open-Source Colloid Simulation Suite	Computers & Chemical Engineering	4.3	Zero-cost, reproducible molecular-to-continuum pipeline	Analytical solutions & Krüss benchmarks

4. Concrete Scilab-Xcos Simulation Prototype

To demonstrate operational feasibility, we present an executable mathematical formulation and Scilab simulation script coupling dynamic interfacial dilation with surfactant diffusion and adsorption kinetics. The governing equations consider an expanding interface with surface excess Gamma(t), bulk subsurface concentration C(0, t), and bulk concentration C0.

The dynamic conservation of surfactant mass at the interface is described by:

d(Gamma)/dt = ja - Gamma * d(ln(A))/dt

where ja is the net adsorption flux given by the kinetic Langmuir-Szyszkowski model, ja = ka * C(0, t) * (Gamma_inf - Gamma) - kd * Gamma, and d(ln(A))/dt represents the dilatational rate imposed by cyclic mechanical expansion (A(t) = A0 * [1 + epsilon * sin(omega * t)]). Under diffusion-limited conditions, the subsurface concentration C(0, t) satisfies the Ward-Tordai convolutional relationship, or in a state-space approximation, a two-compartment boundary layer model with characteristic diffusion boundary thickness delta = (pi * D / omega)^0.5 [36].

The dynamic surface tension gamma(t) is coupled via the Szyszkowski equation of state:

gamma(t) = gamma_0 - n * R * T * Gamma_inf * ln(1 + K_L * C(0, t))

The corresponding Scilab implementation script is displayed below in Code Block 1.

Code Block 1. Scilab implementation of dynamic surface tension under large-amplitude oscillatory area dilation.

```
// Scilab Prototype: Dynamic Surface Tension Under Oscillatory Area Dilation
clear; clc;

// Parameters loaded from SurfactantKit
gamma0   = 72.8;     // Pure solvent surface tension (mN/m)
Gamma_inf= 3.8e-6;   // Maximum surface excess (mol/m^2)
n_ion    = 1.0;      // Nonionic / ionic factor
R_gas    = 8.314;    // Universal gas constant (J/mol/K)
Temp     = 298.15;   // Temperature (K)
ka       = 1.5e-1;   // Adsorption rate constant (m^3/mol/s)
kd       = 1.2e-3;   // Desorption rate constant (1/s)
C0       = 5.0e-3;   // Bulk concentration (mol/m^3)
D_diff   = 4.5e-10;  // Molecular diffusion coefficient (m^2/s)
```

```
// Oscillatory deformation parameters
omega    = 0.5;      // Frequency (rad/s)
eps      = 0.20;     // Large strain amplitude (20%)

// Define state-space derivative function
function dxdt = surfactant_dynamics(t, x)
    Gamma_t = x(1);
    C_sub   = x(2);

    // Dynamic area dilation rate: d(ln A)/dt
    A_t     = 1.0 + eps * sin(omega * t);
    dlnA_dt  = (eps * omega * cos(omega * t)) / A_t;

    // Kinetic adsorption flux
    j_ads    = ka * C_sub * (Gamma_inf - Gamma_t) - kd * Gamma_t;

    // Subsurface boundary-layer mass transfer
    delta    = sqrt(%pi * D_diff / omega);
    j_diff   = (2.0 * D_diff / delta) * (C0 - C_sub);

    // State derivatives: [dGamma/dt; dC_sub/dt]
    dGamma   = j_ads - Gamma_t * dlnA_dt;
    dC_sub   = (j_diff - j_ads) / delta;

    dxdt     = [dGamma; dC_sub];
endfunction

// Time integration grid (0 to 60 seconds)
t = linspace(0, 60, 1200);
x0 = [0.8 * Gamma_inf; C0]; // Initial state

// Execute ODE solver via CVode algorithm
sol = ode(x0, 0, t, surfactant_dynamics);

// Calculate dynamic tension time series
Gamma_res = sol(1, :);
RT_factor = n_ion * R_gas * Temp * Gamma_inf * 1e3; // mN/m
gamma_t   = gamma0 + RT_factor * log(1.0 - (Gamma_res / Gamma_inf));

printf('Simulation completed successfully. Points integrated: %d\n', length(t));
```

5. Strategic Execution Roadmap and Experimental Validation

To translate these theoretical models into published papers in premier Q1 journals, a disciplined four-phase execution roadmap is required:

Phase 1: Verification Against Standard Analytical Solutions (Months 1-2). Benchmark the Scilab-Xcos integrator against pure diffusion-controlled Ward-Tordai solutions for planar interfaces in quiescent solutions, confirming zero numerical drift and convergence of the SUNDIALS CVode algorithm [37].

Phase 2: Validation Using Open Experimental Datasets (Months 3-4). Cross-validate simulated dynamic surface tension curves against established experimental datasets available in literature (e.g., oscillating pendant drop data for sodium dodecyl sulfate [SDS], cetyltrimethylammonium bromide [CTAB], and Triton X-100 measured on Krüss PAT-1M tensiometers) [38].

Phase 3: Identification of Novel Non-Equilibrium Phenomena (Months 5-6). Execute parameter sweeps across mixed surfactant systems, mapping harmonic distortion spectra in dilatational rheology and transient Capillary number thresholds during microfluidic droplet pinch-off.

Phase 4: Manuscript Drafting and Submission (Months 7-8). Package the simulation framework, validated figures, and open-source GitHub repositories according to journal-specific formatting requirements, prioritizing Journal of Colloid and Interface Science and Chemical Engineering Science.

6. Conclusion

The integration of Scilab-Xcos with SurfactantKit resolves a longstanding methodological bottleneck in interfacial engineering. By marrying the microscopic molecular descriptors and thermodynamic rigor of SurfactantKit with the powerful, open-source dynamic continuum solvers of Scilab-Xcos, researchers can explore non-equilibrium interfacial physics that were previously either computationally inaccessible or restricted to costly proprietary software. The four proposed research avenues—spanning nonlinear LAOD rheology, microfluidic tip-streaming, acoustic foam resonance, and

open-source continuum benchmarking—provide a definitive path toward producing premium, peer-reviewed scientific contributions in top-tier colloid and chemical engineering journals.

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